

Robot Control Paradigms

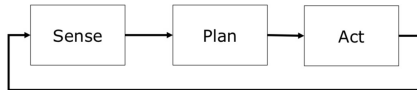
Autonomous Navigation of Mobile Robots

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Classical / Hierarchical Paradigm



- Focus on automated reasoning and knowledge representation
- Perfect world model, closed world assumption

Classical Paradigm



- Stanford AI Laboratory
 - CART, 1973
 - STRIPS (Stanford Research Institute Problem Solver)
- CMU (Moravec)

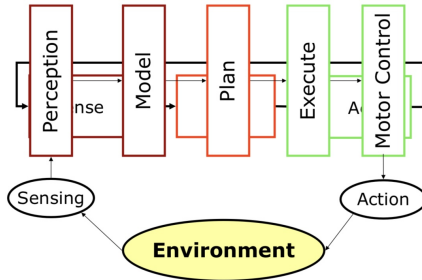
Classical Paradigm

Stanford Cart

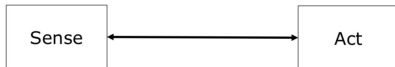
1. Take nine images of the environment, identify interesting points in one image, and use other images to obtain depth estimates
2. Integrate information into global world model
3. Correlate images with previous image set to estimate robot motion
4. On basis of desired motion, estimated motion, and current estimate of environment, determine direction in which to move
5. Execute the motion

Classical Paradigm

as Horizontal / Functional Decomposition

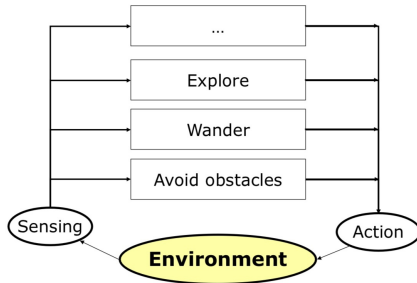


Reactive / Behavior-based Paradigm



- No models – “The world is its own, best model”
- Early successes, but also limitations
- Investigate biological systems

Reactive Paradigm as Vertical Decomposition



Reactive Paradigm

Characteristics

- **Situated** agent, robot is integral part of its environment
- **No memory**, controlled by what is happening in the world
- **Tight coupling** between perception and action via behaviors
- Only local, behavior-specific sensing is permitted (**ego-centric** representation)

Behaviors

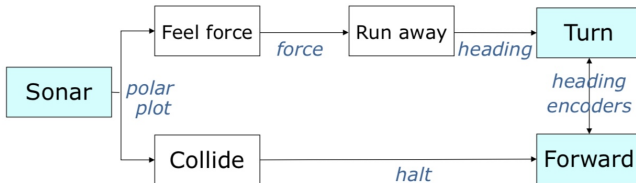
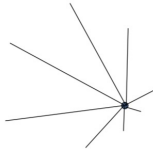
- ...are a **direct mapping** of sensory inputs to a pattern of motor actions that are then used to achieve a task
- ...serve as the basic building blocks for robot actions, and the overall behavior of the robot is **emergent**
- ...support good software design principles due to **modularity**

Subsumption Architecture

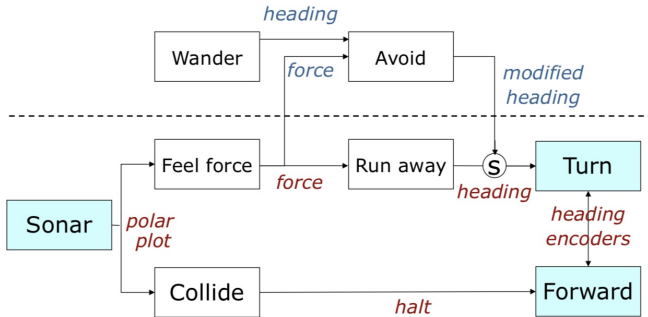
- Introduced by Rodney Brooks in 1986
- Behaviors are networks of sensing and acting modules (augmented finite state machines, AFSM)
- Modules are grouped into layers of competence
- Layers can subsume lower layers
- No internal state!

Level 0: Avoid

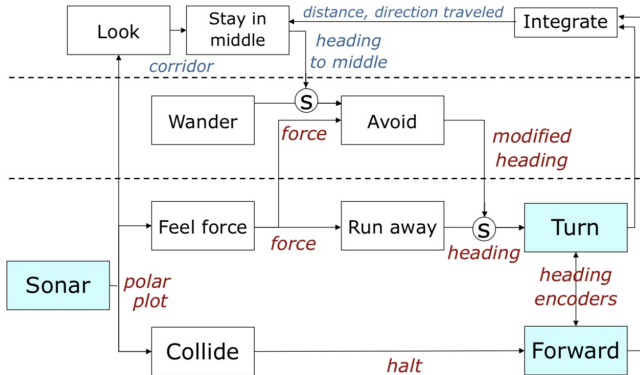
Polar plot of sonars



Level 1: Wander



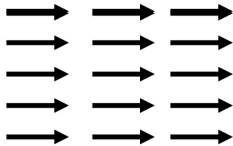
Level 2: Follow Corridor



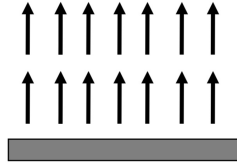
Potential Field Methods

- Treat robot as **particle** acting under the influence of a potential field
- Robot travels along the **derivative of the potential**
- Field depends on obstacles, desired travel directions and targets
- Resulting field (vector) is given by the **summation of primitive fields**
- Strength of field may change with distance to obstacle/target

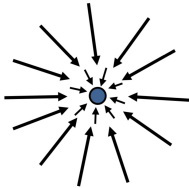
Primitive Potential Fields



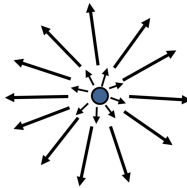
Uniform



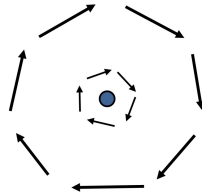
Perpendicular



Attractive



Repulsive



Tangential

Corridor Following

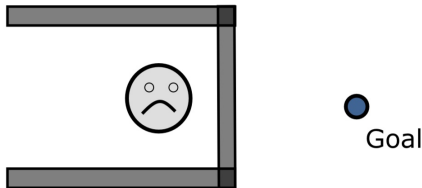
with Potential Fields

- **Level 0** (collision avoidance) is done by the repulsive fields of detected obstacles
- **Level 1** (wander) adds a uniform field
- **Level 2** (corridor following) replaces the wander field by three fields (two perpendicular, one uniform)

Potential Fields

Characteristics

Suffer from **local minima**



- Backtracking
- Random motion to escape local minimum
- Increase potential of visited regions
- Avoid local minima by harmonic functions

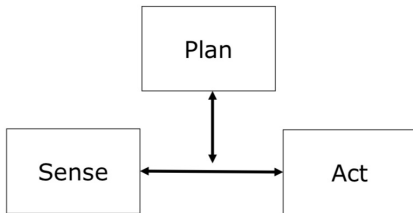
Potential Fields

Characteristics

- No preference among layers
- Easy to visualize
- Easy to combine different fields
- High update rates necessary
- Parameter tuning important

Hybrid Paradigm

Deliberative / Reactive



Combines advantages of previous paradigms

- World model used for planning
- Closed loop, reactive control